

Nanomaterials

Nanomaterials are materials whose characteristic length scale lies within the nanometric range (1-100 nm) at least in one dimension

Classification of Nanomaterials Based on dimension

1. Zero-dimension (0-D) :- Nanomaterials in which all dimensions are measured within the nanoscale. eg: Electron is confined in a 3-D space. Nano particles.
2. One-dimension - Nanomaterials:- are materials in which two dimensions are in nanoscale. eg: Nano wires, Nanofibers, Nanotubes.
3. Two-dimension Nano materials:- are materials in which only one dimension is in nanometric range. eg: Nano films, Nano layers, and Nano coatings.
4. Three-dimension Nanomaterials:- Here all the dimensions are outside the nanometric range. Bundles of nanowires and tubes.

Classification based on Materials

(1) Carbon-based nanomaterials:- These are defined as materials in which the nanocomponent is pure carbon. OR. nanomaterials in which main component is pure carbon
eg: Carbon nano tubes, Fullerenes

(2) Metal based nanomaterials:- Metal based nanomaterials are materials made of metallie nanoparticles like Silver, gold and metal oxides

(3) Nano Composites:- (Composite nanomaterials contain a mixture of simple nanoparticles within a bulk material.) Nano Particles provide better Physical, mechanical and chemical properties to the initial bulk material.

(4) Nano-Polymers or Dendrimers:- Dendrimers are nano sized polymers built from branched units. These are tree like molecules with defined cavities. They can be functionalised at the surface and can hide molecules in their cavities. They are widely used in drug delivery.

Biological Nanomaterials:- These

nanomaterials are biological origin. The important features of these particles are

(1) Self assembly properties

(2) Specific molecular recognition

eg: DNA - nanoparticles

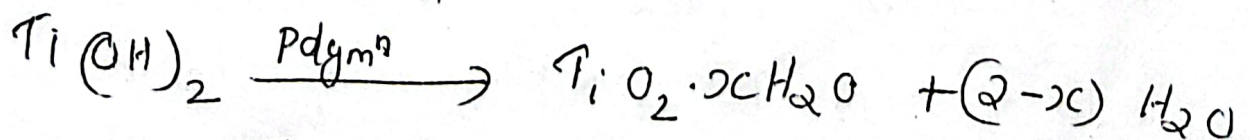
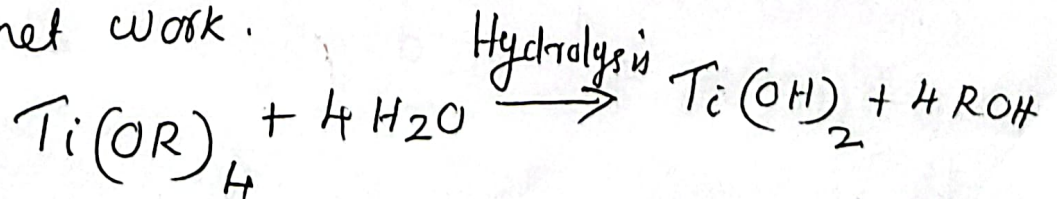
Synthesis of Nanomaterials

(preparation) [chemical methods of preparation]

(1) Hydrolysis Method

Nanoparticles of metal oxides can be prepared by the hydrolysis of their alkoxide solutions under controlled conditions.

eg: Titanium alkoxide get hydrolysed and polymerised to form a three dimensional oxide net work.



The precipitate so obtained was

washed with ethanol and dried at 100°C in a vacuum system for 3 hrs. It is then subjected to annealing process get the nanoparticles.

(2) Sol - Gel Method: is based on the

Phase transformation of a sol into a gel.

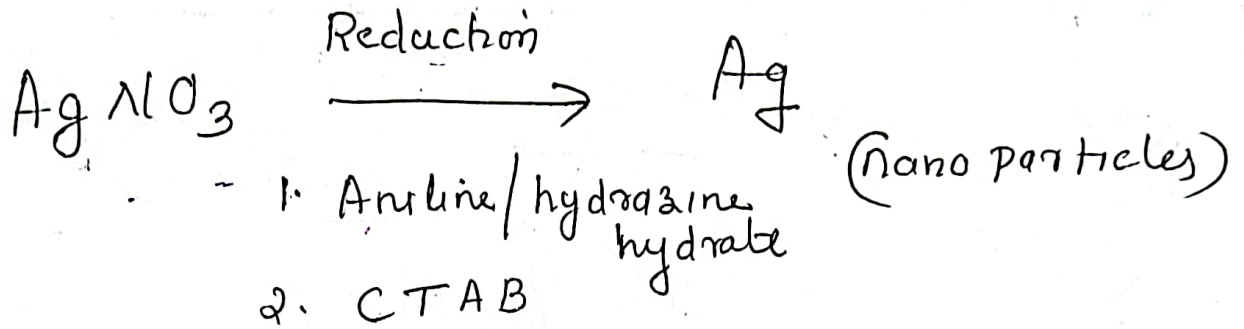
A sol is colloidal system of nano-solid particles dispersed in a liquid. A gel is colloidal system in which liquid droplets are dispersed in a network of solid nanoparticles.

Hydrolysis of metal alkoxides can give a sol at a suitable temperature and pH:

[The sol contains impurities. In order to remove impurities, ~~the sol is the sol~~ the sol is transformed into a gel by changing the pH or other factors. The gel can be purified by filtration and washing with suitable solvents. The purified gel on drying give solid nanoparticles.]

Reduction Nano particles of gold and silver can be prepared by the reduction of their respective solutions using

Reducing agents such as sodium borohydride, hydrazine hydrate, Aniline solution^{etc} along with a protective agent.



Electro reduction

Copper nanoparticles have been prepared by electroreduction process using copper plating bath containing acidified copper sulphate solution. The spongy nanoparticles are formed at the cathode. The spongy layers of copper can be separated to give fine particles.

Applications

- (1) Carbon nanotubes used in transistors.
- (2) Graphene sheets are used as electrodes in ultra capacitors
- (3) carbon nanotube are used for making paper batteries
- ↳ Nano particles are used for the treatment of tumors.